Design	LUTs	Flip-Flops	Latches
Transmitter	33	46	0
Receiver	29	74	0

1. Each board has its own oscillator. Both are nominally 100 MHz but they're never exactly equal, so the phase between them keeps drifting. There's no shared clock, so a signal from one board can change right at an edge of the other board's clock.
2. REQ and ACK come from the other board's clock and can change at any time relative to our clock edges. If one changes close to our edge, it can violate setup/hold and make the first flop metastable. The second flop gives it a full cycle to settle before the logic uses it.
3. The receiver captures DATA after it sees the synchronized REQ change. If DATA changed along with REQ, the receiver could see the new REQ before the new DATA is valid and grab the wrong bit. So DATA is set first and held until ACK comes back, which means the receiver is done with it. DATA doesn't need a synchronizer because it's already stable by the time REQ gets across.
4. SYNC tells the receiver the packet has the format it expects. The checksum only checks that the packet agrees with itself. For example, an all-zero packet passes the checksum (0 ^ 0 = 0) but fails the SYNC check.
5. The XOR checksum works per bit position, so it catches any single-bit flip and most multi-bit errors. It can miss errors that cancel in the same bit position, like the same bit flipped in both the payload and the checksum. It also can't catch a wrong payload that was already wrong before the checksum was computed, and it can't correct anything or request a new payload.
6. Every bit needs a full REQ/ACK round trip: the FSM states, about 2 cycles for REQ to cross, the receiver capturing and sending ACK, then about 2 cycles for ACK to cross back. That's roughly 10 clock cycles per bit instead of 1. Only 8 of the 24 bits in a packet are payload, so the useful data rate is lower again.
7. A parallel bus would send a whole byte per handshake, so there'd be about 8x fewer handshakes and the link would be roughly 8x faster. The cost is 8 Pmod pins and wires instead of 1, but it would get rid of the bit counter and the shifting mux. The flip-flop count only goes up a little.